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Technology-Mediated STEM Interventions for Students with Autism Spectrum Disorder: A Systematic Review and Meta-Analysis

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ABSTRACT

The scientific and technological advances in the 21st century necessitate the acquisition of STEM skills for all students in order to be successful in post-secondary education and the workforce, including students with autism spectrum disorder (ASD). Technology-mediated intervention has shown promise in improving the STEM skills of students with ASD. This meta-analysis aimed to provide a quantitative synthesis of single-case experimental studies of technology-mediated interventions on the STEM achievement of students with ASD. Forty-four studies were included, and the Tau-U effect size (ES) across studies ranged from 0.28 to 1.00, with an omnibus ES of 0.91 (CI_{95} = [0.86, 0.97]). Variables such as technological devices and STEM areas acted as moderators. Implications for research and practice are discussed.

Science, Technology, Engineering, and Mathematics (STEM) education is the integration of one or more of the four disciplines into the education curriculum (Hwang & Taylor, 2016). STEM-related employment opportunities have grown rapidly in the current technologically advanced world. Students with advanced knowledge in STEM are more likely to obtain better job opportunities (Basham & Marino, 2010). The National Science Foundation (2017) highlighted the need for early participation in STEM for all students given the significant impact of STEM skills on everyday life and work. However, STEM educational opportunities have been primarily designed for typically developing students, and students with disabilities have less access to them (Fisher et al., 2022).

STEM interventions for students with ASD

According to the systemizing-empathizing theory, individuals with ASD are below average on empathy but average or above average on systemizing (Baron-Cohen, 2009). Systemizing often requires identification of rules and patterns to analyze and construct systems, which is essential to perform well in STEM-related domains. This suggests that individuals with ASD are likely to be suitable for some STEM occupations (Baron-Cohen et al., 2007). Therefore, it is necessary for instruction to match the needs of students with ASD to improve the accessibility to STEM for diverse learners.

Several studies of individuals with ASD have shown that STEM interventions effectively improved skills in science (e.g., Hart & Whalon, 2012; Knight et al., 2015), technology (e.g., Knight et al., 2019; Spriggs et al., 2015) and mathematics (e.g., Bassette et al., 2019; Yakubova et al., 2015). Ehsan et al. (2018) synthesized 44 studies of STEM interventions for individuals with ASD, including single-case experimental design studies and group design studies. They found that less than half of the studies

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produced positive results ($n = 21$), while others yielded mixed or neutral results. For example, Knight et al. (2015) explored the science intervention through e-books (i.e., Book Builder) to four middle school students with ASD. The results showed that all four students did not show a significant improvement during the Book Builder only intervention. Ehsan et al. (2018) also pointed out that some intervention studies used technology as an assistive tool for teaching STEM content.

Technology-mediated intervention for students with ASD

Over the decades, the use of technology has become increasingly ubiquitous in schools for teaching and learning. In special education settings, the application of technology can relieve learning pressure by (a) providing appropriate support based on the learner's individual needs (Olakanmi et al., 2020), and (b) making learning more engaging and attractive (Knight et al., 2013). Technology refers to an electronic equipment, application, or virtual network that is used to intentionally increase, maintain, and/or improve daily living, work, and leisure capabilities (Odom et al., 2015). Technology could be categories as low technology, moderate technology and high technology (Bouck, 2010). Low technology is commonly referred to a tool that requires little technology, such as tape recorders and calculators. Moderate technology is typically considered as the devices that have electronic components but are not computerized, such as graphing calculators. High technology is commonly referred to the devices that are computerized, such as computers, smartphones, and tablets (Chia et al., 2018). In this review, technology-mediated intervention refers to provision of treatment with the use of high technology to support the process of learning various skills through using computer, tablets, and other mobile devices. Some reviews have noted the benefits and effects of technology-based programs to teach individuals with ASD in various fields including employment skills, social skills and academic skills (e.g., Lee et al., 2018).

More specifically, in STEM education, there are a few reviews providing evidence for the effectiveness of technology use for students with ASD. For example, Wright et al. (2020) provided a narrative review of 10 single-case experimental design studies of the use of video modeling (a type of technical support) in STEM intervention, and found that the use of video modeling was beneficial for providing STEM intervention for students with disabilities. In a more recent review of STEM intervention, Yakubova et al. (2022) examined twenty single-case experimental design studies of STEM interventions for individuals with ASD. They determined that technology-based interventions for teaching STEM skills to K-12 students with ASD can be considered as an established evidence-based practice. However, since these reviews were all descriptive and did not include effect size of each study, the overall effect of integrating technology when teaching STEM content to students with ASD was not quantified and the extent to which contextual variables moderate the effect of STEM interventions was also unknown.

Potential moderating variables

Prior research on STEM intervention or technology-mediated intervention has shed some light on the factors that could moderate the effects of technology. Participant characteristics are nonnegligible when evaluating the effect of a technological intervention. Some analyses indicated that student performance varied according to grade level and participant diagnosis (e.g., Hong et al., 2017). We also aimed to investigate the moderation effects of intervention characteristics. Chauhan (2017) pointed out that different effect sizes might be obtained under different settings (e.g., classroom, home). Besides, some studies indicated that intervention intensity (e.g., duration, frequency), interventionist (e.g., researcher, teacher) and instructional strategy (e.g., explicit instruction, video modeling, manipulative-based instruction) could have impacted the effect of the interventions (e.g., Ok et al., 2020; Peltier et al., 2020).

Moreover, we also paid attention to the target outcomes of an intervention (i.e., target skills), including STEM areas and assessment level. Belland et al. (2017) examined the effects of computer-based

instruction for STEM learners. They categorized assessment level of the target content to concept level, principles level and application level. They found that the effect size for the principles level was significantly higher than the effect size for the concept level. Lastly, we considered moderator variables related to technological characteristics. Ledbetter-Cho et al. (2018) discovered that the effect size of an intervention was significantly different across different operators (i.e., instructor, participant) and that effects also varied based on whether the participants had pre-training on a device. Ok et al. (2020) demonstrated that there was no significant difference between traditional computer devices and mobile devices.

The current study

The primary purpose of this study is to provide a more comprehensive and updated review of the effect of technology-mediated STEM interventions for students with ASD. Given the ubiquity of technology, this study focuses on evaluating the specific effect of technology-mediated STEM interventions that involve computer or mobile devices. The research questions are:

(1) What is the magnitude of the omnibus effect of technology-mediated STEM interventions on students with ASD, as measured by Tau-U?
(2) To what extent, do methodological characteristics, participant characteristics, intervention characteristics, technological characteristics and target skills moderate intervention effect?

Method

Search procedures

A systematic literature search was conducted through four steps. First, we conducted the database search in December 2021 through the Educational Resources Information Clearinghouse (ERIC), Psychological Abstracts (PsycINFO), Web of Science and ProQuest. These databases were searched with no restriction on publication date using the following sets of key terms and roots (combined with 'AND'): (a) autis* or Asperger or ASD; (b) computer* or mobile or laptop or tablet or iPad or touch-screen or PDA or personal digital assistant; (c) STEM or science or technology or engineer* or math*. The search resulted in 5423 studies, 4593 of which were retained after duplicates were removed. Then, we read the titles and abstracts to determine whether the study should be included in our meta-analysis. If the title and abstract cannot determine eligibility, full study was read. In this phase, 31 studies were included. Secondly, we hand searched the 2012–2021 issues of nine journals (i.e., *Journal of Autism and Developmental Disorders*, *Exceptional Children*, *Research in Developmental Disabilities*, *Journal of Special Education*, *Research in Autism Spectrum Disorders*, *Focus on Autism and Other Developmental Disabilities*, *Exceptionality*, *Research and Practice for Persons with Severe Disabilities*, and *Journal of Special Education Technology*), and 5 additional articles were included. Next, we screened the reference lists of three reviews (Ehsan et al., 2018; Wright et al., 2020; Yakubova et al., 2022) related to our topic, and 5 articles were included. Last, we performed ancestral search and citation search of qualified articles through SCOPUS, and three studies were added. Finally, a total of 44 studies were included. The overview of the process of search is shown in Figure 1. The following inclusion criteria were used:

(a) Studies were peer-reviewed articles or dissertations published in English.
(b) Studies utilized single-case research design with a clearly readable graph of data. We chose single-case design studies because compared to randomized controlled trials, single-case design studies can better capture the characteristics of individuals, which are of particular interest to the field of special education (Shadish et al., 2016).

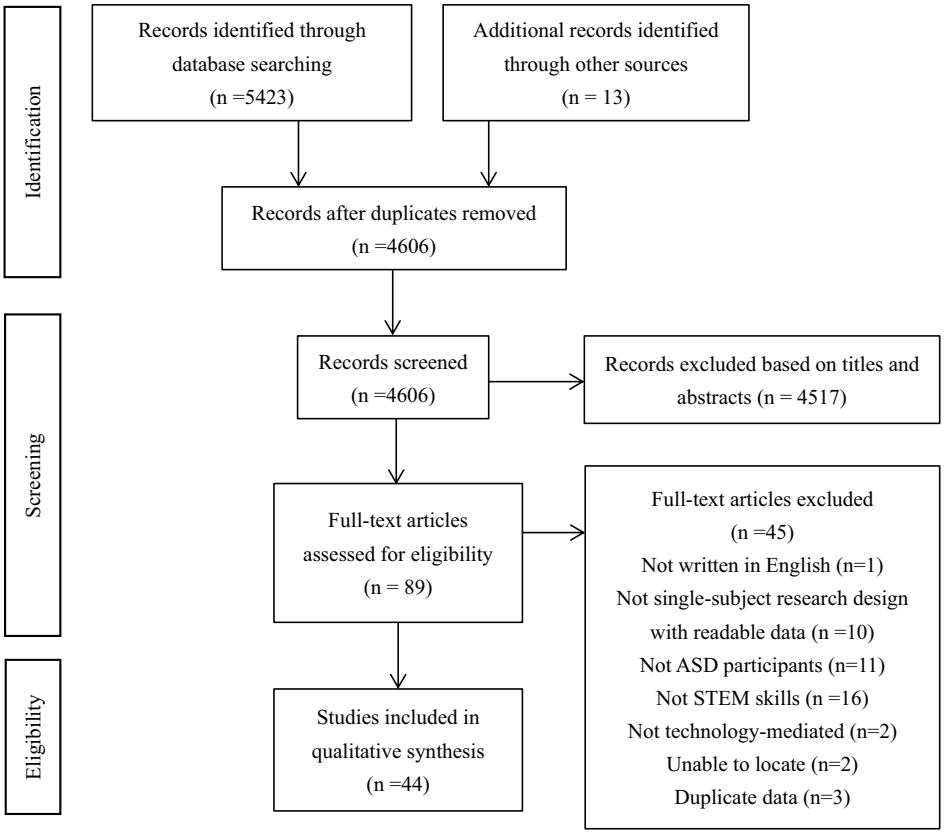


Figure 1. Flowchart for systematic search.

(c) Studies included at least one participant with ASD restricted to school-age and post-secondary students (5–25 years old). Studies that included students with other disabilities, but also included at least one participant diagnosed with ASD were considered acceptable.
(d) Studies implemented technology-mediated intervention which included computer or mobile devices as an intervention component. In addition, the interventions targeted at least one dependent variable related to STEM content.

Quality evaluation

We evaluated the quality of all included studies under the What Works Clearinghouse Standards (WWC, Version 4.1) (Kratochwill et al., 2010). All studies were assessed against the following four aspects: (a) systematically manipulated independent variable, (b) inter-assessor agreement data (IAA), (c) attempts to demonstrate treatment effects, and (d) the number of data points per phase. Each article was rated as 0, 1, 2 points by standard. If a study was rated as 2 points for all the standards, it was classified as 'meeting standards without reservation.' If a study was rated as either 1 or 2 points for all the standards, it was classified as 'meeting standards with reservation.' If a study was rated as 0 points for any standard, it was classified as 'not meeting the standard.'

At first, we conducted a moderator analysis on designs not meeting the standards versus those meeting the standards. Results of this analysis were not statistically significant for all dependent variables. Considering that the meta-analytic approach allows the data to speak for itself (Cooper, 2016), and there is no consensus as to whether studies should be excluded based on design quality prioritized (Morin et al., 2019), we retained studies not meeting the standards. We also conducted

moderator analysis not only within the studies that met the WWC standards but also within all included studies to present comprehensive findings.

In addition, previous research pointed out that the lack of fidelity and social validity evaluation was a shortcoming of the WWC standards (Maggin et al., 2014). Therefore, as Zimmerman et al. (2018) suggested, we supplemented the fidelity and social validity data of included studies to reduce the potential bias of using WWC to evaluate the quality of included studies.

Variable coding and data analysis

Each study was coded for the following categories: (a) methodological characteristics, including procedural fidelity and social validity; (b) participants characteristics, including gender, grade level and comorbid disorder; (c) intervention characteristics, including setting, interventionist, duration and instructional strategy; (d) technological characteristics, including technological device, operator of device and pre-training on device; (e) target skills, including STEM areas and assessment level. Detailed categories and definitions are shown in Appendix A. Comorbid disorder was later dropped in the moderator analysis because the profiles of different participants varied, and therefore it was difficult to include comorbid disorder as a moderator.

The online software WebPlotDigitizer was used to extract the raw graphic data, and an online Tau-U calculator was used to obtain the effect size of each phase contrast. Phase contrast refers to the versus between the baseline phase and the intervention phase. After the effect sizes of all phase contrasts were obtained, we calculated the weighted average of the effect sizes from all contrasts using the inverse-variance weighting method and aggregated the effect size of all the studies (Vannest et al., 2016). Then, we adopted Comprehensive Meta-Analysis Version 3.0 (CMA 3.0) to calculate the omnibus effect size of all included studies and to generate the effect sizes of different subgroups to identify potential moderators. We interpreted the effect sizes as follows: 0–0.62 indicates a small effect; 0.63–0.92 indicates a moderate effect; and 0.93–1.00 indicates a large effect (Parker et al., 2011). Besides, I^2 statistics was used to test heterogeneity across studies (Higgins & Thompson, 2002).

Interrater reliability and publication bias

The first author and the third author worked as raters for each stage of the study independently. Interrater reliability (IRR) was calculated as dividing the number of agreements by the total number of agreements and disagreements and then multiplying by 100. IRR was conducted on 100% of systematic literature search, 100% of quality evaluation, 50% of variable coding and 50% of data extraction. The first author provided a training session for approximately two-hours to the third author using another subset of data before rating. Both authors had experience in searching, rating and coding articles in prior meta-analyses projects. If disagreement occurred, two raters discussed to reach a consensus. IRR results obtained were as follows: 82% for database search, 100% for journal search, 100% for reference search, 100% for ancestral search and citation search, 92% for quality evaluation, 91% for descriptive coding, and 97% for data extraction.

We identified publication bias of included studies by Rosanthal's Fail-Safe N method and the Egger's linear regression method (Borenstein et al., 2009). In the current study, the obtained Fail-Safe N was 1638, which was much larger than the desired Fail-Safe N ($n = 5 \times 44 + 10 = 230$). Egger's test of the intercept (−0.45; CI_{95}= [−1.21, 0.30]; $p = .23$) was non-significant. These results both indicate that publication bias did not exist.

Results

Overall effects and methodological quality

Our systematic search yielded a total of 44 studies, which consisted of 4 dissertations and 40 peer-reviewed articles. Tables 1–3 provide a summary of the information across all included studies. All 44 included studies were published in or after the year of 2012, and the majority of studies (70%) were published in the last 5 years. This trend indicates a rapidly growing interest in technology-mediated interventions for students with ASD in the most recent years. There was a total of 111 participants with 161 phase contrasts across all studies. These studies as a whole fell short of the WWC Standards, with more than half of the studies did not meet the standards ($n = 26$). Only three studies did not report procedural fidelity and social validity, and these studies also did not meet the WWC Standards. Other studies all reported acceptable fidelity value and positive social validity. The Tau-U effect size for all the studies ranged from 0.28 to 1.00, with an overall effect size of 0.91 ($CI_{95} = [0.86, 0.97]$). Among those meeting the WWC Single-Case Design Standards ($n = 18$), the Tau-U effect size ranged from 0.72 to 1.00 with a mean of 0.98 ($CI_{95} = [0.91, 1.00]$). Effect sizes of included studies are reported in Table 1. The I^2 value was 0, which indicated no inconsistencies among these studies.

Moderator analyses

Methodological characteristics, participant characteristics, intervention characteristics, technological characteristics and target skills were examined for their potential influences. Table 4 provides specific results of our analyses under two conditions – when all designs were included (i.e., all) and when only designs meeting the WWC Standards were included (i.e., WWC only). Overall, the results of these analyses were positive with almost all variable categories yielding moderate to large effects (range from 0.719 to 1). The subgroup of studies that did not report procedural fidelity yielded small effects (ES = 0.601, all). Of all the variables, significant differences in effects were detected for the type of technological device ($p = .032$) and the STEM area ($p = .028$), but only when all the included studies were considered. Regarding technological device used, interventions using computers produced moderate effects (ES = 0.741, all), and interventions using mobile devices produced large effects (ES = 0.982, WWC only; ES = 0.931, all). Regarding STEM areas, Tau-U result was the smallest for science, which indicated a moderate effect (ES = 0.731, WWC only; ES = 0.731, all). Tau-U results for other STEM areas indicated large effects: technology (ES = 1.000, WWC only; ES = 0.953, all), mathematics (ES = 0.989, WWC only; ES = 0.944, all) and multiple (ES = 1.000, WWC only; ES = 1.000, all).

Discussion

The current study examined the effects of technology-mediated STEM interventions for students with ASD. The first research question was to examine the magnitude of the omnibus intervention effect. The result revealed that technology-mediated intervention was effective for students with ASD in the STEM domain when computer or mobile devices were used to support intervention. More specifically, most technological software used in the intervention were intended to improve a narrow area of knowledge in a specific STEM area such as mathematics and technology, paired with appropriate instructional strategies. For instance, 'Base Ten Blocks' can illustrate numerical concepts in a tangible way and it can be used as virtual manipulative to teach place value and computational problems (e.g., Bassette et al., 2019). 'OzoBlockly digital programming' that allows for programming of block-based codes can be used to teach students how to program robots digitally in the framework of explicit instruction (e.g., Knight et al., 2019).

We also aimed to identify potential moderators. In particular, methodological characteristics, participant characteristics, and intervention characteristics did not show statistically significant moderator effects. The result can be interpreted as positive because technology-mediated STEM

Table 1. Methodological characteristics and participant characteristics of studies.

	Methodological characteristics				Participant characteristics			
Study name	SCED	WWC	Procedural fidelity	Social validity	Gender	Grade level	Comorbid disorder	Tau-U
Bassette et al. (2019)	AT	2	Yes	Yes	*M* = 3	E	Speech and language impairment	1.00
Bassette et al. (2020)	AT	2	Yes	Yes	*M* = 3	E	Language impairment	1.00
Beckman et al. (2019)	RE	0	No	No	*M* = 1	E	Fragile X syndrome	0.92
Bouck et al. (2014)	AT	0	Yes	Yes	*M* = 3	E	NR	1.00
Bouck et al. (2019)	MP	1	Yes	Yes	F = 1	Mi	NR	1.00
Bouck, Maher, et al. (2020)	MP	1	Yes	Yes	F = 1	Mi	NR	1.00
Bouck, Park, et al. (2020)	MP	0	Yes	Yes	F = 1	Mi	NR	1.00
Bouck, Shurr, et al. (2020)	MP	0	Yes	Yes	F = 1	Mi	NR	0.98
Bouck and Park (2020)	MP	1	Yes	Yes	F = 1	Mi	NR	1.00
Bouck, Park, et al. (2020)	MP	0	Yes	Yes	*M* = 1 F = 1	Mi	NR	1.00
Bouck et al. (2021)	MP	0	Yes	Yes	*M* = 1 F = 1	Mi	NR	0.92
Burton et al. (2013)	MP	0	Yes	Yes	*M* = 3	Mi	Intellectual disability	1.00
Gilley et al. (2021)	MP	1	Yes	Yes	F = 1	P	NR	1.00
Hamdi (2020)	MB	2	Yes	Yes	*M* = 4	E	NR	0.98
Hart and Whalon (2012)	RE	2	Yes	Yes	*M* = 1	H	Intellectual disability, hearing loss, speech and language impairment	0.72
Howorth and Raimondi (2019)	MP	0	Yes	Yes	*M* = 3	Mi	Speech and language impairment	0.61
Howorth (2015)	MP	0	Yes	Yes	*M* = 6	Mi	Speech and language impairment	0.47
Jimenez and Besaw (2020)	MB	0	Yes	Yes	*M* = 2	E	Intellectual disability	0.88
Jowett et al. (2012)	MB	1	Yes	Yes	*M* = 1	K	NR	1.00
Knight et al. (2015)	MP	0	Yes	Yes	*M* = 3 F = 1	Mi	Intellectual disability	0.65
Knight et al. (2018)	MP	0	Yes	Yes	*M* = 1 F = 1	E	Intellectual disability	0.98
Knight et al. (2019)	MP	0	Yes	Yes	*M* = 2 F = 1	H	Emotional and behavior disorder	1.00
Langhorne (2017)	MB	0	No	Yes	*M* = 3	H	NR	0.57
Ledbetter-Cho et al. (2020)	MP	1	Yes	Yes	*M* = 5	E	Intellectual disability, speech impairment	1.00
McKissick et al. (2018)	MP	0	Yes	Yes	*M* = 2 F = 1	Mi	Intellectual disability	0.70
McMahon et al. (2016)	MP	0	Yes	Yes	*M* = 1	P	NR	0.97
Morris et al. (2021)	MB	1	Yes	Yes	F = 1	Mi	ADHD	1.00
Park et al. (2020)	MP	1	Yes	Yes	*M* = 2 F = 1	Mi = 2 E = 1	NR	1.00
Park et al. (2021)	MP	1	Yes	Yes	*M* = 1	E	NR	1.00
Rezowalli (2017)	RE	0	No	No	*M* = 1	Mi	NR	0.28
Root et al. (2017)	MP	0	Yes	Yes	*M* = 3	E	Intellectual disability	1.00
Root and Browder (2019)	MP	0	Yes	Yes	F = 3	Mi	Intellectual disability	0.91

(*Continued*)

Table 1. (Continued).

	Methodological characteristics				Participant characteristics			
Study name	SCED	WWC	Procedural fidelity	Social validity	Gender	Grade level	Comorbid disorder	Tau-U
Root et al. (2019)	MP	2	Yes	Yes	*M* = 2 F = 1	E	ADHD, intellectual disability, language impairment	0.96
Root et al. (2020)	MP	0	Yes	Yes	*M* = 2	Mi	Language impairment	0.94
Root et al. (2021)	MP	0	Yes	Yes	*M* = 2 F = 2	P	NR	1.00
Shurr et al. (2021)	AT	0	Yes	Yes	*M* = 3	E	NR	0.82
B. R. Smith et al. (2013)	MP	0	Yes	Yes	*M* = 3	Mi	ADHD, intellectual disability	1.00
K. A. Smith et al. (2015)	MP	0	Yes	No	*M* = 3	H	Speech and language impairment	0.80
Spriggs et al. (2015)	MP	1	Yes	Yes	*M* = 3 F = 1	H	ADHD, epilepsy, sensory integration disorder	1.00
Weng and Bouck (2014)	MP	0	Yes	Yes	*M* = 3	H = 1 Mi = 2	Intellectual disability	0.41
Wright et al. (2019)	MP	0	Yes	Yes	*M* = 3	Mi	Intellectual disability	1.00
Yakubova et al. (2015)	MP	1	Yes	Yes	*M* = 3	H	NR	1.00
Yakubova et al. (2016)	MB	2	Yes	Yes	*M* = 3 F = 1	K = 3 E = 1	Intellectual disability	0.97
Yakubova et al. (2020)	MP	1	Yes	Yes	*M* = 2 F = 1	Mi	ADHD, epilepsy, auditory processing disorder, specific learning disability	1.00

SCED=Single case experiment design, MP = Multiple probe design, MB = Multiple baseline design, AT = Alternating treatment design, RE = Reversal design, M = Male, F = Female, K = Kindergarten, E = Elementary, Mi = Middle school, H = High school, P = Postsecondary, NR = Not reported.

interventions are effective regardless of methodological characteristics, participant characteristics and intervention characteristics. In other words, although STEM areas tend to be challenging for students with ASD, incorporation of technology has shown to be a promising support (Basham & Marino, 2010). As for instructional strategies, video modeling and manipulative-based instruction have been used most widely. Additionally, it is worth noting that although some interventions employed the same instructional strategy, their intervention protocol varied. Therefore, it is difficult to determine whether the characteristics of the strategy or the intensity of the intervention exerted a larger impact on the overall effect (Lee et al., 2018).

Characteristics of technology used is a critical factor to consider. Technology device as a moderator was found to be statistically significant. Technology devices were divided into computers and mobile devices (e.g., iPad and Samsung tablet). The interventions using mobile devices showed larger effects than those using computers, which was inconsistent with Ok et al. (2020). One possible explanation is the different number of devices used in the included studies. Ok et al. identified far more studies using computer ($n = 16$) interventions than using tablet ($n = 4$), while we identified far more studies using

Table 2. Intervention characteristics and target skills of studies.

	Intervention characteristics				Target skills	
Study name	Interventionist	Setting	Total duration (min)	Instructional strategy	STEM areas	Assessment level
Bassette et al. (2019)	Researcher	MS	NA	MBI	Mathematics	Pr
Bassette et al. (2020)	Researcher	MS	NA	MBI	Mathematics	C, Pr
Beckman et al. (2019)	Teacher	S	180	VM	Mathematics	A
Bouck et al. (2014)	Therapist	O	NA	MBI	Mathematics	Pr
Bouck et al. (2019)	Researcher	S	120	MBI	Mathematics	C
Bouck, Maher, et al. (2020)	Researcher	S	160	MBI	Mathematics	C, Pr
Bouck, Park, et al. (2020)	Researcher	S	100	MBI	Mathematics	Pr
Bouck, Shurr, et al. (2020)	Researcher	S	175	MBI	Mathematics	Pr
Bouck and Park (2020)	Researcher	S	28	MBI	Mathematics	Pr
Bouck, Park, et al. (2020)	Researcher	S	75	MBI	Mathematics	Pr
Bouck et al. (2021)	Researcher	S	NA	MBI	Mathematics	Pr
Burton et al. (2013)	Researcher Teacher	S	100-200	VM	Mathematics	A
Gilley et al. (2021)	Researcher	S	50-100	SBI	Mathematics	A
Hamdi (2020)	Researcher	S	225-360	VM	Mathematics	Pr
Hart and Whalon (2012)	Researcher Teacher	S	650	VM	Science	C
Howorth and Raimondi (2019)	Researcher	S	225-275	EI	Science	C, A
Howorth (2015)	Researcher	S	NA	EI	Science	C, A
Jimenez and Besaw (2020)	Teacher	S	NA	MBI	Mathematics	Pr
Jowett et al. (2012)	Researcher	O	NA	VM	Mathematics	C
Knight et al. (2015)	Researcher	S	130-310	EI	Science	C, A
Knight et al. (2018)	Researcher	G	NA	VM	Mathematics Science	C
Knight et al. (2019)	Researcher	S	NA	EI	Technology	A
Langhorne (2017)	Researcher	NR	180-210	MBI	Mathematics	A
Ledbetter-Cho et al. (2020)	Teacher	S	8-60	VM	Mathematics	Pr
McKissick et al. (2018)	Researcher	S	120-357	EI	Science	C
McMahon et al. (2016)	Researcher	G	48-75	AR	Science	C
Morris et al. (2021)	Researcher	S	105-360	VM	Mathematics	Pr
Park et al. (2020)	Researcher	S	NA	EI	Mathematics	Pr
Park et al. (2021)	Researcher	S	NA	EI	Mathematics	Pr
Rezowalli (2017)	Researcher	S	50-75	VM	Mathematics	Pr
Root et al. (2017)	Researcher	G	60-120	SBI	Mathematics	A
Root and Browder (2019)	Researcher	G	70-135	SBI	Mathematics	A
Root et al. (2019)	Researcher	O	75	SBI	Mathematics	A
Root et al. (2020)	Researcher Teacher	G	165	EI	Mathematics	A
Root et al. (2021)	Researcher	G	75-300	AR	Mathematics	A
Shurr et al. (2021)	Researcher	S	NA	VM	Mathematics	Pr
B. R. Smith et al. (2013)	Researcher	G	NA	EI	Science	C, A
K. A. Smith et al. (2015)	Teacher	O	NA	VM	Technology	A
Spriggs et al. (2015)	Teacher	S	NA	VM	Mathematics Technology	A
Weng and Bouck (2014)	Researcher	S	NA	VM	Mathematics	A
Wright et al. (2019)	Teacher	S	NA	VM	Technology	A
Yakubova et al. (2015)	Researcher	S	NA	VM	Mathematics	A
Yakubova et al. (2016)	Researcher	S	240-330	VM	Mathematics	Pr
Yakubova et al. (2020)	Researcher	S	NA	VM	Mathematics	Pr

G = General education setting, S = Special education setting, O = Other setting, MS = Multiple settings, NA = Not available,EI = Explicit Instruction, VM = Video Modeling, SBI = Schema-based Instruction, MBI = Manipulative-based Instruction, AR = Augmented Reality-based Instruction, C = Concept, Pr = Principles, A = Application.

Table 3. Technology matrix of studies.

Study name	Hardware used	Software used	Pre-training	Operator
Bassette et al. (2019)	iPad	Base Ten Blocks app	Yes	Participant
Bassette et al. (2020)	iPad	Fraction Tiles app	Yes	Participant
Beckman et al. (2019)	Samsung Galaxy 5.0 tablet	I-Connect application	Yes	Participant
Bouck et al. (2014)	desktop computer	National Library of Virtual Manipulatives	Yes	Participant
Bouck et al. (2019)	iPad	fraction manipulative app	No	Participant
Bouck, Maher, et al. (2020)	iPad	fraction manipulative app	No	Participant
Bouck, Park, et al. (2020)	iPad	Cuisenaire® Rods app	No	Participant
Bouck, Shurr, et al. (2020)	NA	Color Tiles app	No	Participant
Bouck and Park (2020)	iPad	Two color counters app	Yes	Participant
Bouck, Park, et al. (2020)	iPad	Cuisenaire® Rods app	No	Participant
Bouck et al. (2021)	iPad	Number Lines app	Yes	Participant
Burton et al. (2013)	iPad	Panasonic HM-TA1 flip video camcorder and Windows Live Movie Maker software	No	Participant
Gilley et al. (2021)	iPad air	the Go Worksheet app and a calculator	No	Participant
Hamdi (2020)	iPad	iMovie film-editing programme	Yes	Participant
Hart and Whalon (2012)	iPad	NA	No	Instructor
Howorth and Raimondi (2019)	iPad	the iAnnotate application	No	Participant
Howorth (2015)	iPad	digital text and annotation software	Yes	Participant
Jimenez and Besaw (2020)	iPad	GoWorksheet Maker application	No	Participant
Jowett et al. (2012)	iPad	iMovie film-editing programme	No	Instructor
Knight et al. (2015)	computer	Book Builder™	Yes	Participant
Knight et al. (2018)	Samsung tablet	VivaVideo tablet application	No	Participant
Knight et al. (2019)	iPad	OzoBlockly digital programming	No	Participant
Langhorne (2017)	computer	i-Ready™	Yes	Participant
Ledbetter-Cho et al. (2020)	iPad air	PowerPoint embed video models	Yes	Participant
McKissick et al. (2018)	desktop computer	Slide	No	Participant
McMahon et al. (2016)	iPad	Aurasma app	Yes	Participant
Morris et al. (2021)	iPad	The HP Reveal app	Yes	Participant
Park et al. (2020)	iPad	Base 10 Blocks app	No	Participant
Park et al. (2021)	iPad	Color Tiles app	No	Participant
Rezowalli (2017)	iPad	iMovie application	No	Participant
Root et al. (2017)	iPad	The SMART notebook© application	Yes	Participant
Root and Browder (2019)	iPad	The SMART notebook© application	No	Participant
Root et al. (2019)	iPad air	the GoWorksheet app	Yes	Participant
Root et al. (2020)	iPad	The SMART notebook© application	No	Participant
Root et al. (2021)	iPod touch	calculator app, HP Reveal app, and First-Then Visual Schedule HD app	Yes	Participant
Shurr et al. (2021)	iPad	the 'Base Ten Blocks' app	Yes	Participant
B. R. Smith et al. (2013)	iPad	Keynote® software	Yes	Participant
K. A. Smith et al. (2015)	mobile device	NA	Yes	Participant
Spriggs et al. (2015)	iPad	My Pictures Talk™ application	Yes	Participant
Weng and Bouck (2014)	iPad	iMovie application	No	Participant
Wright et al. (2019)	iPad, Samsung Galaxy 4-8	OzoBlockly digital programming	Yes	Participant
Yakubova et al. (2015)	iPad	NA	No	Participant
Yakubova et al. (2016)	iPad	NA	No	Participant
Yakubova et al. (2020)	iPad	NA	No	Participant

NA = not available.

Table 4. Effect sizes by potential moderator.

	Designs meeting WWC			All designs		
Variables	Group Tau-U[CI_{95}]	Cases	*p*	Group Tau-U[CI_{95}]	Cases	*p*
Methodological characteristics						
Procedural Fidelity			\			.051
Yes	0.982[0.911,1]	69		0.921[0.869,0.973]	155	
No	\	0		0.601[0.284,0.918]	6	
Social Validity			\			.229
Yes	0.982[0.911,1]	69		0.918[0.865,0.970]	155	
No	\	0		0.728[0.422,1]	6	
Participant characteristics						
Gender			.900			.259
Male	0.980[0.900,1]	55		0.898[0.840,0.955]	132	
Female	0.991[0.835,1]	14		0.971[0.858,1]	29	
Grade			.968			.211
Preschool	0.987[0.864,1]	16		0.987[0.864,1]	16	
Elementary	0.985[0.862,1]	29		0.953[0.862,1]	58	
Middle school	1.000[0.838,1]	15		0.835[0.746,0.924]	57	
High school	0.937[0.741,1]	9		0.874[0.724,1]	19	
Postsecondary		0		0.987[0.778,1]	11	
Intervention characteristics						
Setting			.973			.264
General education	1.000[0.745,1]	8		0.976[0.856,1]	35	
Special education	0.976[0.889,1]	47		0.885[0.821,0.950]	100	
Others	0.992[0.852,1]	14		0.973[0.852,1]	23	
Interventionists			.716			.910
Researcher	0.988[0.909,1]	55		0.905[0.846,0.964]	119	
Teacher	1.000[0.796,1]	9		0.939[0.812,1]	29	
Therapist	\	0		1.000[0.684,1]	6	
Multiple	0.869[0.584,1]	5		0.896[0.676,1]	7	
Duration			.795			.344
0–2 hours	0.989[0.786,1]	12		0.957[0.827,1]	33	
Over 2 hours	0.958[0.842,1]	24		0.881[0.793,0.969]	45	
Instructional strategies			.995			.208
Explicit instruction	1.000[0.725,1]	4		0.778[0.655,0.902]	28	
Video modeling	0.978[0.896,1]	47		0.945[0.871,1]	68	
Schema-based	0.956[0.547,1]	3		0.955[0.744,1]	13	
Manipulative-based	1.000[0.813,1]	15		0.918[0.810,1]	42	
AR-based instruction	\	0		0.985[0.765,1]	10	
Technological characteristics						
Technological devices			\			.032*
Mobile devices	0.982[0.911,1]	69		0.931[0.877,0.985]	145	
Computers	\	0		0.741[0.575,0.906]	16	
Pre-training			.844			.741
No	0.977[0.889,1]	40		0.921[0.851,0.990]	72	
Yes	0.992[0.870,1]	29		0.903[0.827,0.964]	89	
Operator			.740			.512
Instructor	0.960[0.810,1]	9		0.960[0.810,1]	9	
Participant	0.989[0.908,1]	60		0.906[0.852,0.961]	152	
Target skills						
STEM areas			.736			.028*
Science	0.731[0.286,1]	2		0.731[0.601,0.860]	29	
Technology	1.000[0.611,1]	2		0.953[0.770,1]	17	
Math	0.989[0.914,1]	63		0.944[0.885,1]	113	
Multiple	1.000[0.639,1]	2		1.000[0.639,1]	2	
Assessment level			.994			.077
Concept	0.965[0.825,1]	12		0.933[0.818,1]	24	
Principles	0.987[0.893,1]	43		0.954[0.876,1]	66	
Application	0.989[0.786,1]	11		0.907[0.808,1]	52	
Multiple	1.000[0.677,1]	3		0.719[0.559,0.879]	19	

*$p < .05$, **$p < .01$, ***$p < .001$.

tablet ($n = 40$) interventions than using computer ($n = 4$) in the current study. Besides, tablet devices have become more popular recently, and they are more mobile than computers; therefore, mobile devices allow students with ASD to receive an intervention in a setting they are familiar with (Knight et al., 2013). In addition, the availability of inexpensive apps in tablets provides convenient and flexible features for users such as touching screen, using stylus, and tilting the device to interact with the trigger. Computers are usually only accessible via keyboard and mouse. However, only 16 contrasts from 4 studies (10%) involved using computers in mathematics and science, and none of them met the WWC Standards. Therefore, the result should be interpreted with caution. As for operator and pre-training, the results showed no significant differences between subgroups, which was contrary to Ledbetter-Cho et al. (2018). One possible explanation is that with the popularization of technology, participants are familiar with the use of these devices in their daily life and have no problems using them, so who operates the device or whether this person has pre-training does not matter.

Considering target skills, only STEM area was found to have significantly different effects among subgroups. In particular, interventions on science skills showed the smallest effects as compared to those on other STEM disciplines which was inconsistent with Chauhan (2017). Chauhan (2017) discovered that the use of technology was more effective for science than for mathematics. One possible explanation was that participants in Chauhan's study were typically developing elementary students, while participants in current study were students with ASD. The interventions on science skills focused on the concept level or both the concept level and the application level, and these levels are too abstract and difficult for students with ASD to master in a short time (Taylor et al., 2020). However, given the lack of replication studies that taught the same content, we need to be cautious when generalizing this finding. In terms of number of studies, only two focused on multiple disciplines. Although two STEM disciplines were included in these two studies, the effects on the two disciplines were assessed separately. Of the remaining 42 studies, the majority of them focused on mathematics skills (e.g., basic numeracy, arithmetic operations, fractions and geometry); seven focused on science skills (e.g., animals, natural resources and habitats); three focused on technology skills (e.g., data entry and coding); no relevant studies focused on engineering skills. It is understandable since mathematics is a core discipline and the importance of mathematics interventions for individuals with ASD have been underlined (Gevarter et al., 2016). However, technology-based interventions, as a valuable asset, are under-utilized in other three STEM areas.

Limitations and future directions

There are some limitations of the current study that need to be noted. Firstly, although we did not exclude gray literature during screening, only four dissertations without peer-reviewed process were included. The omnibus effect size is likely inflated because positive findings are more likely to be published (Hopewell et al., 2009). Our results should be interpreted with caution, and more studies should be conducted to verify our results.

Secondly, the current analysis was limited by the omission of participant functioning levels and intervention intensity data, such as frequency, in many of the studies reviewed. Since they were important factors that often impact the effectiveness of an intervention (Aspiranti et al., 2020; Olakanmi et al., 2020), we recommend that future research provide more specific and detailed information about demographic characteristics and intervention characteristics. This will help to clarify the applicability of their intervention and facilitate the design of the intervention (Lee et al., 2018).

Thirdly, we focused on the use of personal computers and mobile devices in STEM interventions for students with ASD. Future research could explore the impact and effectiveness of other types of technology (e.g., virtual reality) in supporting STEM students with ASD.

Finally, we used the WWC standards to evaluate the quality of our included studies, which only captured the quality of the study design without consideration of additional important factors related to the internal validity of the included studies. Future researchers may add additional evaluation standards specific to special education research such as CEC quality indicators.

Implications for practice

The current review provides three implications for helping practitioners design and deliver intervention to students with ASD. Firstly, there is significant evidence to encourage the use of computer, tablets, and other mobile devices when teaching STEM contents for students with ASD. There are data to support the use of technology to teach mathematics, the acquisition of science, technology, and engineering skills using technology has yet to be sufficiently examined. Meanwhile, we should recognize that the wide variability in daily special education classrooms is much different from that in highly controlled studies setting, and we cannot exaggerate the role of technology without consideration of the appropriate instructional strategies (Knight et al., 2013). Specifically, the most used strategy is explicit instruction for teaching science, and video modeling for teaching technology. As for mathematics, multiple strategies are broadly used such as manipulative-based instruction, schema-based instruction as well as explicit instruction and video modeling.

Secondly, practitioners need to choose appropriate technology that match both student's needs and the STEM content. It is required to investigate if the technology could be an effective learning tool for each student based on the student's characteristics (e.g., needs, ability, preference). In addition, practitioners should consider which apps could be effective for teaching STEM target skills (Ok et al., 2016).

Thirdly, teachers can collaborate with researchers in special education to promote a wider range of practices, considering that both teachers and researchers can be effective for implementing STEM interventions. We can offer more educational opportunities to students with ASD and promote equity in education through collaboration (Spooner et al., 2011).

In conclusion, the purpose of this meta-analysis was to examine the effect of technology-mediated STEM interventions for students with ASD. Technology-mediated STEM interventions can be effective in teaching students with ASD across methodological characteristics, participant characteristics and intervention characteristics. However, we focused on the use of personal computers and mobile devices. And the majority of studies aimed at mathematics skills and no relevant studies focused on engineering skills. Future reviews could explore the effectiveness of other types of technology in supporting STEM students with ASD.

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